Joshua Trujillo

Mechanical Engineering
2027

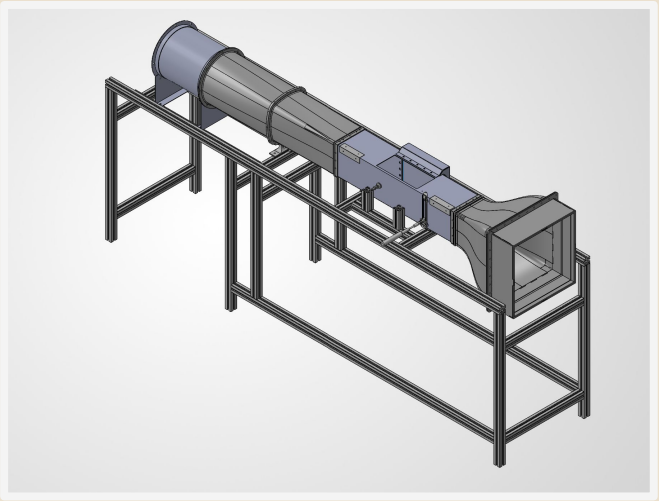
Updated as of Aug 2026

Engineering Portfolio

Low Speed Wind Tunnel

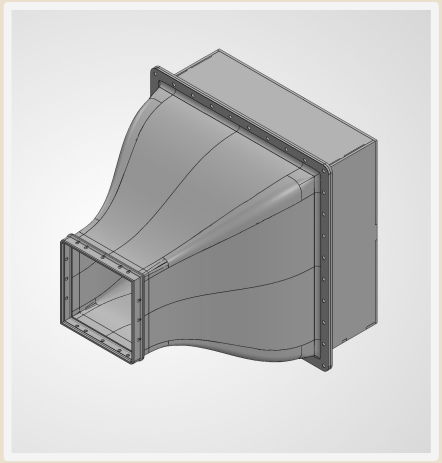
May 2026 - Present

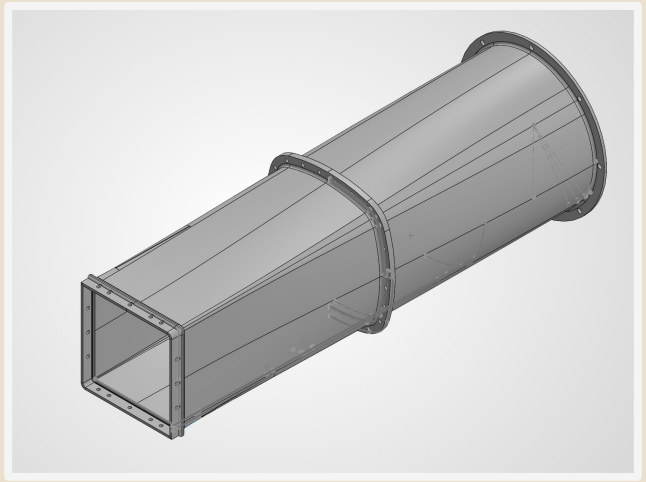
- Developing a low-speed wind tunnel to investigate how airflow influences heat transfer, supporting research on fire behavior and battery fires.
- The design integrates a contraction nozzle, test section, downstream ducting, and fan assembly on a structural frame to provide controlled airflow through the experimental region.
- Designed and fabricated a 3D-printed contraction nozzle to transition airflow into the test section, refining the connection to improve fit, alignment, and sealing.
- Created custom support brackets in SolidWorks to secure and align tunnel components, incorporating 3D printing to iterate on the design during assembly.

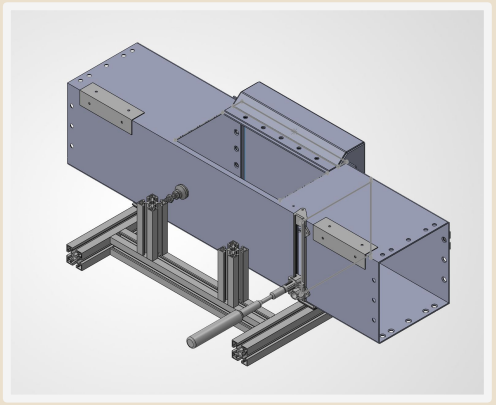


Low Speed Wind Tunnel

May 2026 - Present



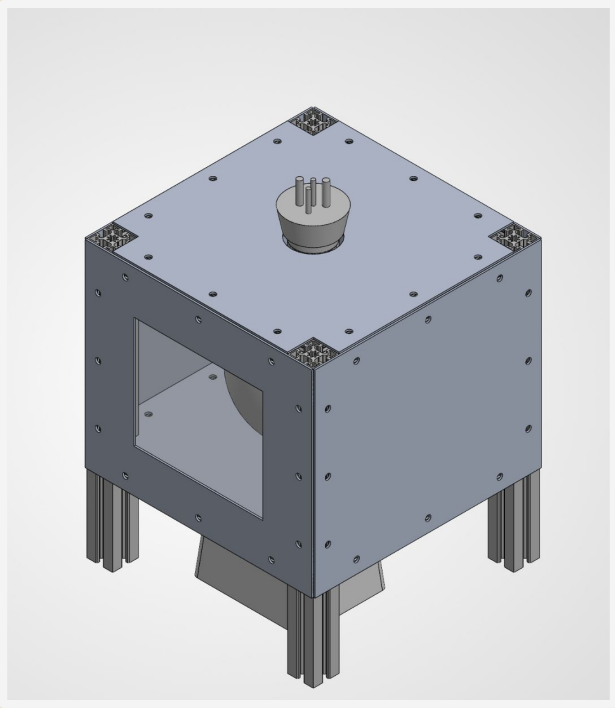




ASTM E681 Flammability Limit Test Apparatus

May 2026 - Present

- Designed a insulated flammability chamber to ASTM E681 spec, with an engineered explosion-relief vent sized for safe deflagration venting.
- Built a thermocouple closed-loop temperature control system (MAX31856 + SSR) with a electrical safety interlock chain.
- Designed a 4-line gas/vacuum/pressure manifold enabling real-time pressure monitoring during fill and evacuation — a departure from the standard single-line approach.



Jan 2026 - May 2026

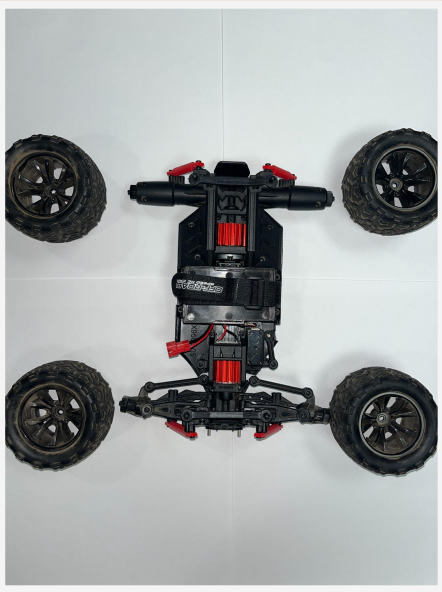
RC car Project

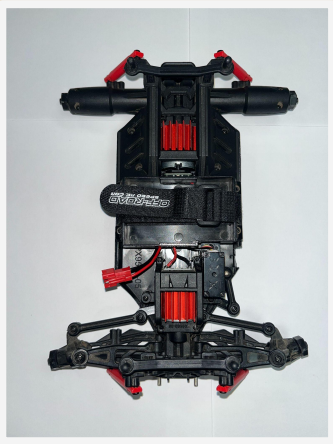
- Designed and optimized an RC car for competitive racing, iterating on steering, suspension, and drivetrain systems to improve speed, handling, and overall cornering performance.
- Redesigned and modified an RC car to improve its performance across multiple racing events, focusing on balancing speed, maneuverability, and overall handling.
- Improved the steering system and suspension geometry, reducing the turning radius from 28 inches to 20 inches for tighter cornering and better control.
- Built, tested, and adjusted the vehicle’s steering, suspension, and chassis setup based on its performance during different race conditions







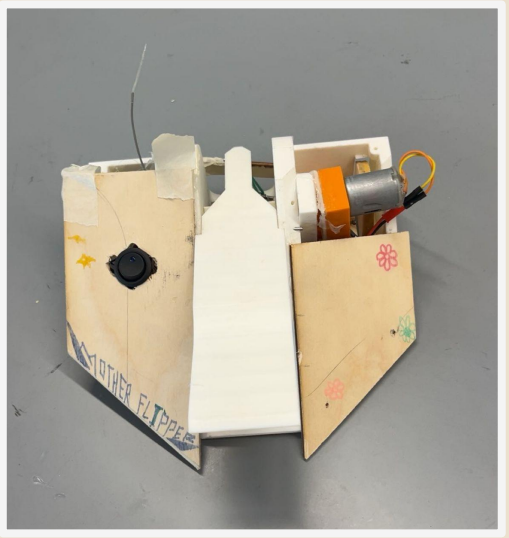


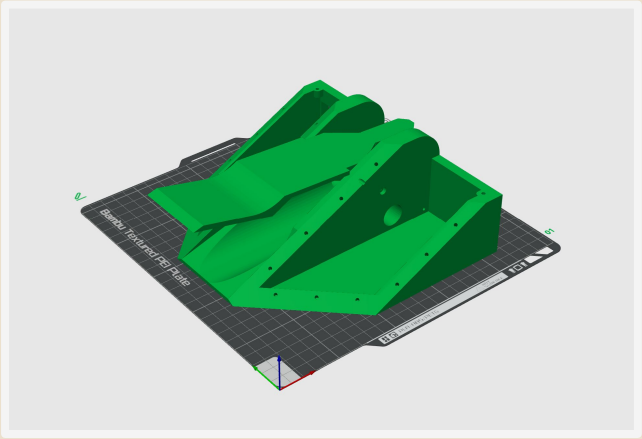


Jan 2025 - May 2025

Battle Bot Design Project

- Collaborated on a four-person team to design and build a 1 lb flipper-style battle bot for a Mechanical Design course.
- Co-designed the robot body in SolidWorks, balancing weight, structural durability, and internal component packaging.
- Helped develop the electrical system and integrate a high-RPM motor, gearbox, springs, and cam-driven flipper mechanism.




Bambu Textured PEI Plate

